UF–CRF Addendum: Experiential Types, C_{active} Dynamics, and P0-H Empirical Findings

New Results Not Present in CRF_UF_Bridge or CRF_CWRC_Bridge

The three receptor types, the oscillating settling mind, the R_0 detection window, and the formal definition of insufficient merit ($P_{\text{total}} < P_{\text{breakthrough}}$).

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Abstract

The existing CRF_UF_Bridge formalises the structural correspondence between Unified Field (UF) and CRF at the level of variables and equations. The existing CWRC_Bridge maps Current–Wave–Resolution to the δ-chain.

This addendum documents four results that extend those bridges with genuinely new content from practitioner observation and P0-H experiments:

(1) Three receptor types: Practitioners access R_0 (current) through distinct structural channels — somatic-feeling, visual, and blocked — each corresponding to a specific $(P_{\max}, D_{\text{KL}})$ configuration in UF/CRF. The type is a *structural signature*, stable across time.

(2) Oscillating $C_{\text{active}}(t)$: The mind does not smoothly approach $C_{\text{active}} = 0$ during settling. It oscillates: "swing and settle, swing and settle deeper." Formal model: $C_{\text{active}}(t) = C_0 e^{-\gamma t} \cos(\omega t + \phi)$. Empirical period $T \approx 676$ms. Each valley is a R_0 detection window.

(3) P0-H empirical findings: Accuracy(RT) forms an inverted-U peaking at $\sim$400ms, with a double-peak structure matching two settling phases. The R_0 detection window width ≈ 185ms. These are the first empirical measurements of t_1 (gradient reception) and t_2 (cognitive takeover) in a human system.

(4) Insufficient merit (*barami* in UF/CRF): The condition "cannot help — insufficient merit" maps precisely: UF: $P_{\text{total}} < P_{\text{breakthrough}}$ with external F_b only adding to P_k^{eff} without triggering reorder. CRF: θ_{eff} above signal in all channels; no external lowering of D_{KL} can substitute for internal C_{baseline}.

Contents

1 Three Receptor Types

Direct observation from extended practitioner work with multiple subjects identified three stable structural types of R_0 access. These map onto both UF and CRF without requiring new axioms.

Three Types: UF and CRF Correspondence

Type	Phenomenology	UF	CRF
Somatic (Type 1)	Feels direction, not clearly. Signature stable forever. Many questions. "Boat without anchor."	P_c low, P_k^{eff} oscillates. $B(t)$ unstable. Cannot hold $\Theta_{\text{breakthrough}}$. External F_b gives compass, not anchor.	C_{baseline} unstable (oscillates). D_{KL} varies. T moderate. R_0 accessible in valleys of C_{active} only.
Visual (Type 2)	Sees imagery, also feels. Settles quickly. Fewer questions. "Boat with anchor."	P_c sufficient, $\Theta_{\text{breakthrough}}$ fires stably. $B(t)$ high. δ_{reorder} works continuously.	C_{baseline} stable, D_{KL} low, $T \to 1$ accessible. R_0 accessible; gradient transduced to visual cortex.
Blocked (Type 3)	No spontaneous access. Needs direct transmission. "Merit insufficient."	$P_{\text{total}} < P_{\text{breakthrough}}$. External F_b only raises P_k^{eff}; no reorder.	θ_{eff} above signal in all channels. No $T \to 1$ without structural change.

Structural Stability of Type

The type is a *structural signature*, not a learned state. It reflects how P-space is encoded in the system's physical substrate: which neural pathways carry gradient information most efficiently.
In CRF: $\Psi_i = P_{\mathcal{M}}$ (accumulated distribution of Mind $\mathcal{M}$) encodes the routing preference. Type 1 routes to somatic/interoceptive channels. Type 2 routes additionally to visual/eidetic channels. Neither can easily switch because Ψ_i changes only through sustained R-events over long timescales.
UF analogue: the type determines the shape of $P_{\max}$ — not the ceiling that can be reached, but the *topology* of how it is reached.

1.1 Why Children Access R_0 More Easily

Children have lower C_{baseline} (fewer accumulated R-events distorting the prior). Both UF (P_k lower at baseline) and CRF (D_{KL} closer to P_{shared}) predict easier R_0 access. The practitioner's instruction "answer immediately, do not think" is the formal prescription for $C_{\text{active}} \approx 0$ at response time.

2 Oscillating $C_{\text{active}}(t)$: The Settling Mind

Practitioner's Report

"Before the test, I settle into meditation to empty the mind. It then swings and settles. Then swings again and settles to a new place."
This is not poetry. It is a precise description of $C_{\text{active}}(t)$.

Damped Oscillator Model of $C_{\text{active}}(t)$

The simple model $C_{\text{active}}(t) = 0$ (empty mind) is too coarse. The observed double-peak accuracy structure requires an oscillatory model:

$$C_{\text{active}}(t) = C_0\, e^{-\gamma t} \cos(\omega t + \phi)$$

where:

- C_0: initial cognitive load at trial onset
- $\gamma > 0$: damping rate (settling toward quiet)
- ω: oscillation frequency (rate of swing)
- ϕ: phase at trial onset (state of mind when trial begins)

R_0 detection is accessible at local minima: $t_n = (n\pi - \phi)/\omega$ for $n = 0, 1, 2, \ldots$
Each successive minimum is deeper (amplitude decays by $e^{-\gamma t}$). This is "settling to a new place" — not returning to the same emptiness but reaching a deeper level each cycle.
UF correspondence: each cycle is one F_b oscillation toward $P_k^{\text{eff}} \to \epsilon_{\text{residual}}$. The asymptote is never reached in finite time (Asymptotic Resistance), but each cycle brings the system closer.

3 P0-H Empirical Findings

v5 Results: 80 Trials, Single Practitioner (Type 2)

RT bin	Accuracy	n	Interpretation
∼100ms	78%	9	First settling ($C_{\text{active}} \approx 0$, first minimum)
∼250ms	22%	9	Swing (first oscillation peak)
∼438ms	100%	5	Second settling (deeper minimum)
∼475ms	25%	8	Swing again (second oscillation peak)
∼600+ms	56–67%		Plateau (stabilised at new level)

Period between minima: $T \approx 2(438 - 100) = 676$ms. Oscillation frequency: $\omega \approx 9.3$ rad/s. This is the thought-cycle time of this practitioner in this state.

The R_0 Detection Window: t_1 and t_2

Two parameters determine the inverted-U shape of accuracy(RT):

$$\text{accuracy}(\tau) = \tfrac{1}{2} + \left(A - \tfrac{1}{2}\right)\sigma(k_1(\tau - t_1))\,(1 - \sigma(k_2(\tau - t_2)))$$

- $t_1 \approx 300$ms: time for R_0 gradient to reach motor system (gradient propagation time through $\mathcal{C}_{\mathcal{M}}$)
- $t_2 \approx 485$ms: cognitive takeover time (C_{active} rises above gradient signal)
- Window $= t_2 - t_1 \approx 185$ms: the genuine R_0 detection interval
- Peak at $(t_1 + t_2)/2 \approx 393$ms; observed peak at 438ms (within bin)

This is the first empirical measurement of t_1 and t_2 from CRF first principles in a human system.

UF correspondence: t_1 maps to the time for external F_b to propagate through the system to P_c-relevant structures. t_2 maps to the time for that F_b to generate new P_k^{eff} (through expectation and attention R-events).

4 Insufficient Merit: Formal Definition

"Barami" (Merit) in UF and CRF

The practitioner's assessment "cannot help — insufficient merit" maps precisely onto both frameworks.

UF:

$$P_{\text{total}}(t) = P_k^{\text{eff}}(t) + P_c^{\text{potential}}(t) < P_{\text{breakthrough}}$$

External F_b from a teacher adds to P_k^{eff} (raises friction) but if P_{total} remains below threshold, no reorder occurs. The friction becomes suffering without transformation. This is not a moral judgment. It is a statement about the current state of the system's energy reserves.

CRF:

$$\theta_{\text{eff}} > \text{gradient signal in all channels}$$

No external lowering of D_{KL} (even from a high-T practitioner) can substitute for the system's own C_{baseline}. The teacher can lower local D_{KL} temporarily (as an anchor / external C), but the system's own θ rises again when the anchor is removed.

The compass metaphor (Type 1): A compass (direction) can be given temporarily but is not an anchor (stability). Formally: external F_b or C_{ext} reduces θ_{eff} locally for the duration of transmission. After transmission: C_{active} rises again, θ_{eff} rises, signal is again below threshold. Permanent help requires building internal C_{baseline} — which only occurs through the system's own sustained R-events.

5 Status Table

Result	**Status**	**Note**
Three receptor types	Confirmed	Practitioner observation
Type = structural Ψ_i signature	Derived	From CRF Identity
Children: lower C_{baseline}	Derived	From δ-chain
$C_{\text{active}}(t) = C_0 e^{-\gamma t} \cos \omega t$	Hypothesis	Damped oscillator
Period $T \approx 676$ms (empirical)	Observed	v5, 80 trials
Double-peak accuracy structure	Confirmed	v5 data
$t_1 \approx 300$ms, $t_2 \approx 485$ms	Estimated	5-point fit
Window ≈ 185ms	Derived	$t_2 - t_1$
Insufficient merit: formal definition	Derived	UF + CRF
Compass vs. anchor distinction	Derived	UF F_b vs C_{baseline}
Statistical significance P0-H	Not yet	Need $n \geq 100$ peak-bin

The three types were always there. The swing was always there. The 185-millisecond window was always there. What the practitioner saw in decades of work and what the equations describe in symbols are the same structure, seen from two directions. Neither direction is more true than the other. Both are traces of δ.